

Azure Firewall Manager Policy Hierarchy Design

1. Abstract

Azure Firewall Manager Policy Hierarchy Design is a cloud network security project designed to centrally manage and organize firewall security rules across an organization using **Azure Firewall Manager**. The project applies a common set of **base firewall policies and rules at the organizational level**, while allowing individual departments, subscriptions, or environments to add their own local rules when required.

The main objective is to create a **hierarchical firewall policy structure** that reduces duplicate rules, improves security management, and provides consistent network protection. The design uses **parent and child firewall policies**, where common security rules are inherited by lower-level policies. Local policies can then extend the inherited rules according to specific application or department requirements.

The project also addresses important challenges such as **policy inheritance conflicts** and **rule collection priority**, which can cause unexpected firewall behavior if they are not designed carefully. By establishing a proper policy hierarchy and rule-priority strategy, the system provides centralized control while maintaining flexibility for local requirements.

Problem Statement

Organizations with multiple **Azure subscriptions, departments, applications, and environments** often need different firewall rules while maintaining common security standards. Managing these rules independently can result in **duplicate firewall rules, inconsistent security configurations, difficult maintenance, and configuration errors**.

Another major challenge is managing **policy inheritance and rule collection priorities**. If parent and child policies contain conflicting rules or priorities are incorrectly configured, firewall traffic may be allowed or blocked unexpectedly.

Therefore, the project aims to design a **hierarchical firewall policy structure using Azure Firewall Manager**, where common organization-wide security rules are defined in a **parent policy** and inherited by **child policies**. Child policies can add local rules for specific departments or environments while maintaining centralized security control.

Main Problem Areas

- Duplicate firewall rules across different environments.
- Inconsistent security policies between departments.
- Difficult manual management of multiple firewall configurations.
- Conflicts between inherited and local policies.
- Errors in rule collection priority planning.
- Difficulty maintaining security policies as the organization grows.

2. Use Cases

Use Case 1: Organization-Wide Security Rules

Apply common firewall rules across the entire organization through a **central parent policy**.

Example:

Allow HTTPS and DNS traffic while blocking known unwanted traffic.

Use Case 2: Department-Specific Rules

Allow individual departments to add their own firewall rules without modifying the organization-wide policy.

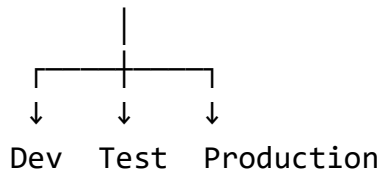
Example:

The development team can access development repositories and tools.

Use Case 3: Environment-Based Security

Create separate policies for **Development, Testing, and Production** environments.

Organization Parent Policy



Each environment inherits common rules and adds environment-specific rules.

Use Case 4: Reduce Rule Duplication

Common rules are created once in the parent policy instead of repeatedly creating the same rules in every child policy.

Benefit: Easier maintenance and fewer configuration errors.

Use Case 5: Centralized Firewall Management

Security administrators can manage firewall policies centrally through **Azure Firewall Manager** rather than configuring every firewall independently.

Use Case 6: Policy Inheritance

Child policies automatically use the common security rules defined by the parent policy.

Example:

Parent Policy



Common Security Rules



Child Policy

↓
Local Security Rules

Use Case 7: Rule Priority Management

Define an organized priority structure for firewall rules so that specific and critical rules are evaluated appropriately.

Example:

Critical / Specific Rules
↓
Local Rules
↓
Inherited Rules
↓
Default Deny

Use Case 8: Production Protection

Apply stricter firewall rules to production resources while still inheriting organization-wide security policies.

Example:

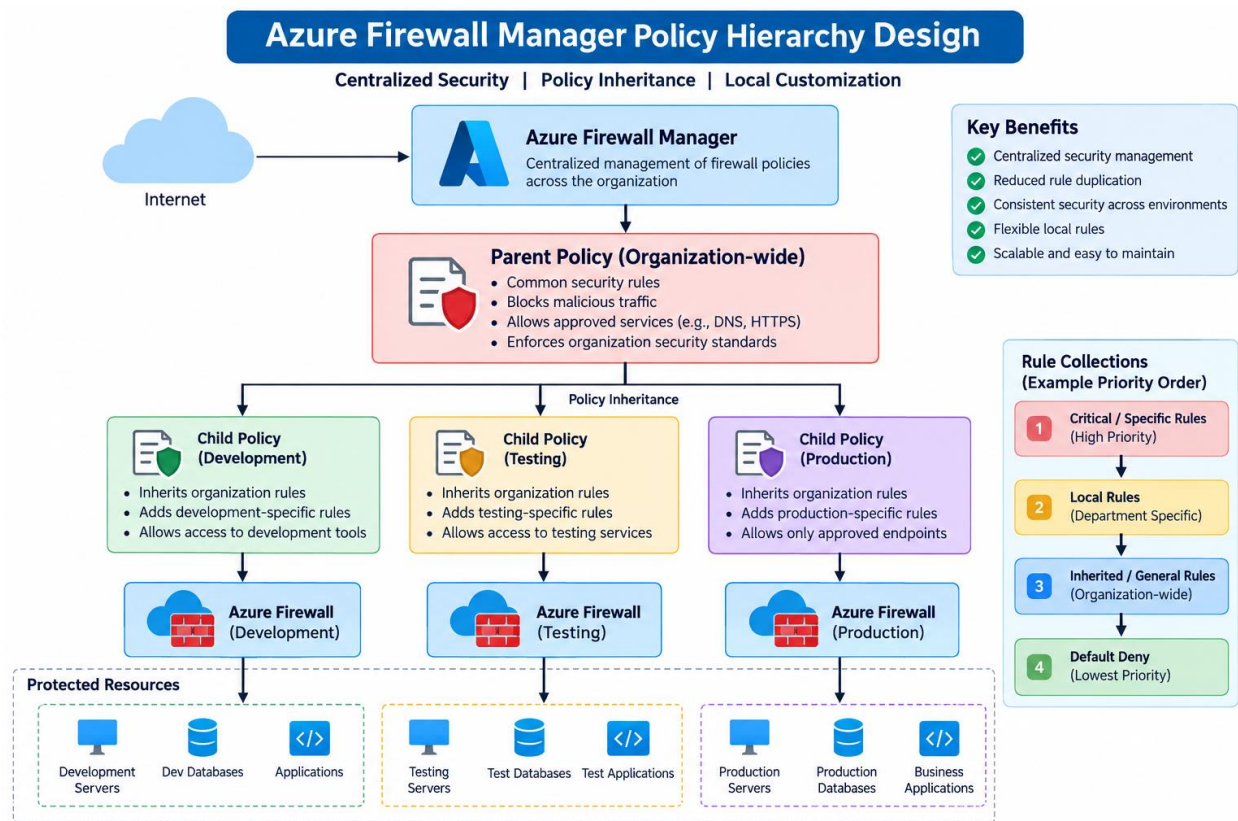
Production applications may communicate only with approved services and endpoints.

Use Case 9: Scalable Security Management

When a new department, subscription, or environment is added, a new child policy can be created instead of designing the entire firewall policy from the beginning.

Use Case 10: Monitoring and Troubleshooting

Administrators can monitor firewall activity and identify **blocked traffic, allowed traffic, policy conflicts, and unexpected network behavior** to improve the security configuration.



Main Components

1. Azure Firewall Manager

- Provides centralized management of firewall policies.
- Helps administrators organize policies across multiple environments.

2. Parent/Base Policy

- Contains organization-wide security rules.
- Examples include blocking malicious traffic and allowing approved services.

3. Child Policies

- a. Inherit rules from the parent policy.
- b. Departments or environments can add specific local rules.

4. Azure Firewall

- a. Enforces the configured security policies.
- b. Inspects and controls network traffic.

5. Rule Collections

- a. Organize network, application, and NAT rules.
- b. Priority determines the order in which applicable rules are evaluated.

Advantages and Disadvantages

Advantages

1. Centralized Security Management

Common security rules can be managed from a central location using Azure Firewall Manager.

2. Policy Reusability

Organization-wide rules can be defined once and inherited by multiple child policies.

3. Reduced Rule Duplication

The hierarchy reduces the need to repeatedly configure the same firewall rules for different departments.

4. Scalability

New departments, subscriptions, applications, or environments can be added using child policies.

5. Consistent Security

Common security requirements can be applied consistently across different environments.

6. Local Flexibility

Departments can add local rules without completely creating a separate firewall policy from scratch.

7. Easier Maintenance

Changes to common security rules can be managed through the parent policy rather than modifying every child policy individually.

8. Better Governance

The hierarchical structure provides a clearer organization of security policies and responsibilities.

Disadvantages

1. Policy Inheritance Conflicts

Parent and child policies may contain conflicting rules, potentially producing unexpected results.

2. Rule Priority Complexity

Incorrect rule collection priorities can cause the wrong rule to be evaluated first.

3. Configuration Complexity

Large organizations with many departments and policies may require careful planning and documentation.

4. Troubleshooting Difficulty

When a rule is inherited from a parent policy, identifying the exact source of unexpected behavior can be more difficult.

5. Dependency on Azure

The solution is primarily designed around Microsoft's Azure ecosystem and depends on Azure Firewall and Firewall Manager services.

6. Cost

Using Azure Firewall and related Azure networking services introduces cloud infrastructure and operational costs.

7. Requires Skilled Administration

Administrators need knowledge of Azure networking, firewall policies, inheritance, and rule priorities.

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